HTT observational applications: conditional results from retained products

HTT observational research

14 October 2026

Abstract

We executed retained-product observational diagnostics, a selected 24-group CF4 scenario posterior and latent predictive image, and a joint Union3–DESI compressed distance posterior. Scientific interpretation remains conditional on the declared processing, selection, covariance and model assumptions. No physical shear, global tilt, native Bianchi family, or physical MES admission follows.

1 Joint distance result

Union3 has 22 compressed nodes, not 22 independent supernovae. One DESI BGS $q_{\rm iso}$ summary enters once. In the flat late-time matter–Lambda scenario, $E(z)^2 = \Omega_m(1+z)^3 + 1 - \Omega_m$ and $I(z) = \int_0^z du/E(u)$. We analytically integrate a common magnitude offset $M \sim N(43, 5^2)$. Proper priors are $\Omega_m \sim U(.01, .99)$ and $H_0 r_d \sim U(5000, 20000)$ km/s. The reference factorization scenario gives $\Omega_m = 0.357456 \pm 0.027084$ and $H_0 r_d = 9974.63 \pm 204.18$ km/s. It does not separate H_0 and r_d. Five covariance scenarios $\rho = 0, \pm.25, \pm.5$ preserve the released marginal covariances. These are assumptions, not measured cross-covariances or posterior odds. Orders128,256,512 close the declared quadrature criteria. The posterior image retains all original 262144 node IDs and weights. Physical MES fields remain unavailable, with unknown mass 1; imposed distance geometry is not measured zero anisotropy.

2 Selected CF4 scenario

For 24 SNIa-only grouped observations, $x = \log d_A$, $t = c\log(1+z_{\rm true})$, $\mu = 25 + (5/\ln 10)x + 2(5/\ln 10)t/c + \eta$. A single shared zero point is integrated with the preselection law $L = [\int p(\eta)\prod_i f_i d\eta]/[\int p(\eta)\prod_i Z_i d\eta]$. The finite-box endpoint posterior has 2048 nodes and ESS1120.83. The endpoint coefficient is not cosmological H_0. Same-state $d_L = d_A(1+z_{\rm true})^2$ and repeated-measurement PPC give exceedance 0.678020; this is not a uniform null p-value. Thirty-two frozen same-law injections resolve; endpoint-H bias -0.357125 and RMSE 1.953703. The interval study is finite conditional Gaussian-Wald calibration, not empirical CF4 coverage. Population, velocity width, calibration and remainder are scenarios. No certified remainder upper bound is obtained.

3 Temperature and polarization

SMICA tensor-family conditional rank is17/241; a joint 48-choice processing scan has rank 63/241. Both procedures reuse the same sky and the same 300 paired simulations. Each has disjoint training 60 and calibration 240 subsets, but the two results are not independent. No improvement relative to historical p=.039 is inferred. Temperature quadrupole is not physical shear, and residual ell1 is not a physical dipole. BK18's actual 66 correlatedBB spectra in 9 bins give fixed-fiducial HL descriptor $Q_{\rm HL} = 545.104273$, with independent/native algebra parity. It is not a PTE, normalized evidence or primordial-r fit. CLASS 200 unique-per-split noise pairs produce a six-feature joint conditional rank 0.00621118; EE drives this maximum, whereas the displayed BB components are near their noise means. Its independent split-noise law is an approximation; signal leakage/systematics are not calibrated. The first raw FFP10 pair closes six-transform spin2 linearity and signed cross-term checks under two formal U encodings. Raw-product frame/convention/processing binding remains HOLD; no ensemble or actual-data inference is made.

4 Other retained products and limitations

The ACT–Planck released joint lensing amplitude, its Planck marginal and the Planck PR3 reference are overlapping products; they are not multiplied independently. The source-qualified Planck PR3 corrected reference amplitude is $A_{PP} = 1.009562 \pm 0.026253$. Its reference cosmology already includes Planck lensing; this is release consistency, not independent calibration. CMB marginalization has used primary CMB information and does not establish product independence. Source-matched Planck/ACT/SPT spectrum residuals retain their reference, covariance and foreground/calibration scope. BeyondPlanck mean maps are not independent skies. COSMOS scans all released 784016 rows, including zero zfinal placeholders; retention is not completeness or a galaxy-population law. Gaia fits 4756 selected frame sources with 2x2 covariance blocks; upstream outcome selection prevents an independent physical frame test. JWST/CCHP photometry, KiDS n(z), Union3 deletion and SDSS observer-box controls retain their shared-source and conditional assumptions.

5 Statistics versus deterministic error budgets

For the same4756 Gaia sources, thin block whitening gives a6 by9512 influence operator and sampling covariance MCM^T. Changing numerical fitting precision normalization alone leaves this covariance unchanged when sampling C is fixed. For a TOTAL whitened bias norm ρ, coordinate support is $B_j = \rho\sqrt{V_{jj}}$. For fixed raw tangent-coordinate box limits, $B_j = \bar{b}\sum_k |M_{jk}|$. We retain statistical reference halfwidth $1.96\sigma_j$, deterministic support and their sum separately. The box is basis dependent; each coefficient can have a different attaining pattern. These are coordinatewise working-normal sensitivities, not simultaneous or empirical coverage. Endogenous selection and unmeasured cross-source covariance remain limitations. No physical error budget or CAS07 Jacobi remainder is certified; the new Lean wrapper is UNVERIFIED.

6 Provenance and review

Exact generating result files, code/input identities, fixed assumptions, figures and review scopes accompany this report in report.json and figure_inventory.json. Requested author/reviewer settings are separate from observed runtime UNKNOWN. Prior scoped results and upper proofs are reused. Physical joint-model and external native atlas dependencies remain gated. See the Korean report and RETURN for exact missing inputs and resume commands. Publication is not scientific admission.

Primary source context. Union3 release; Planck component separation; Planck lensing; ACT lensing software; BICEP/Keck XIII; CLASS release; COSMOS-Web; Gaia celestial frame.

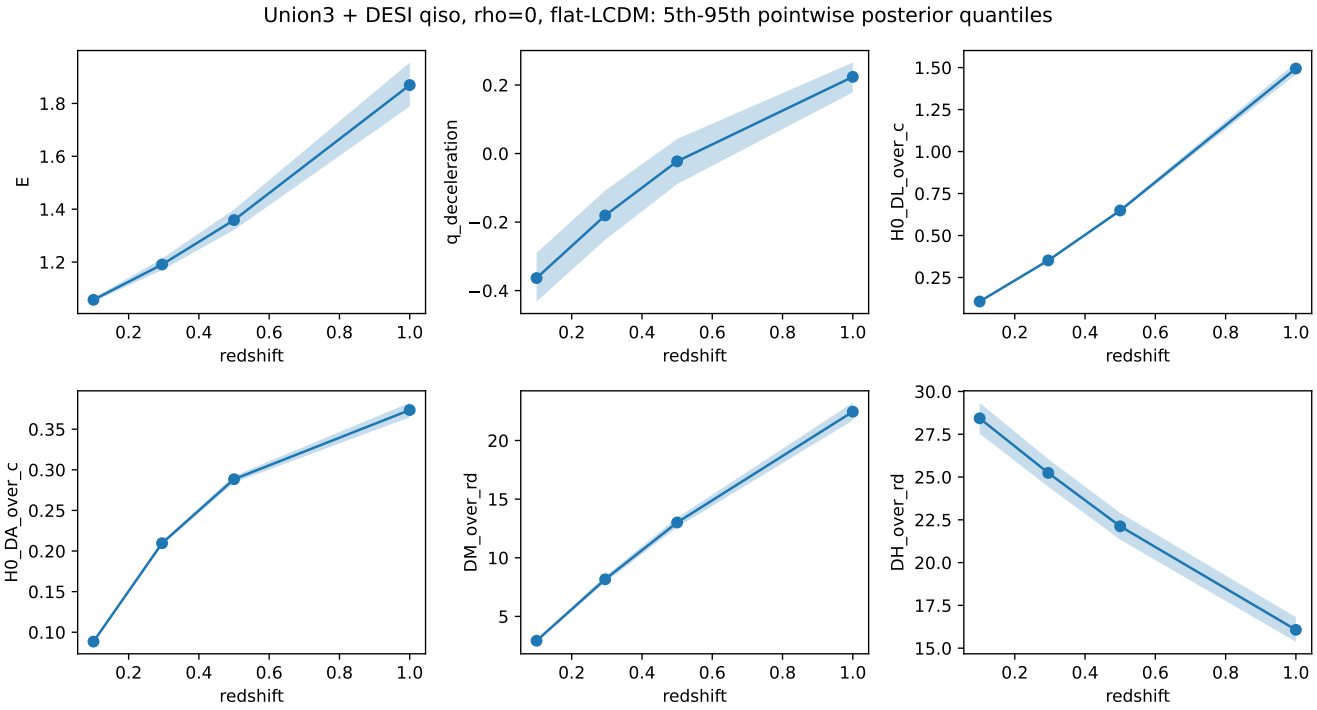


Figure 1: Same posterior rho=0 flat-LCDM distance functions; pointwise5/95% quantiles; physical MES unavailable

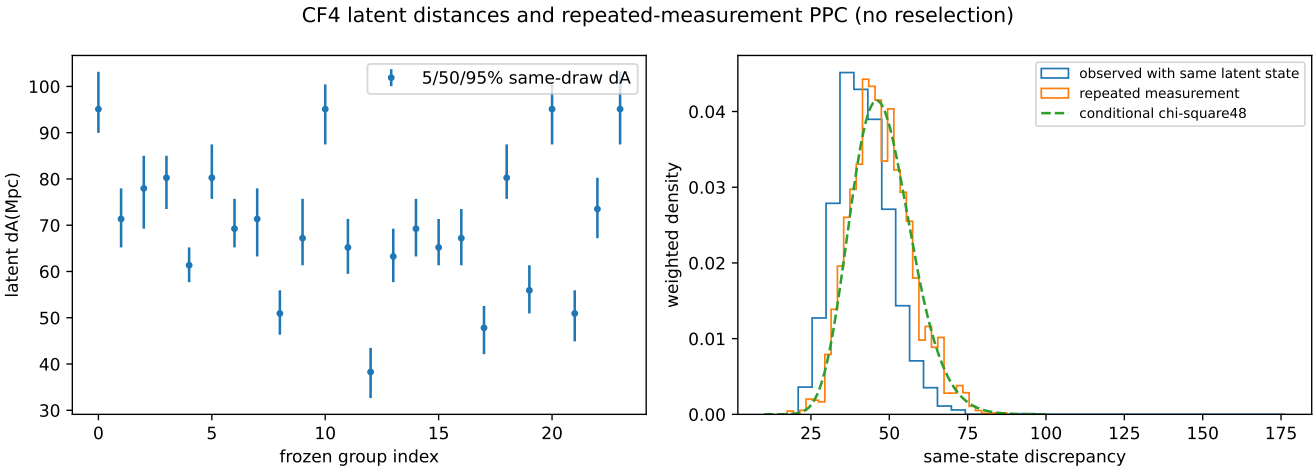


Figure 2: Same posterior latent dA and repeated measurement PPC, without reselection; conditional selected scenario

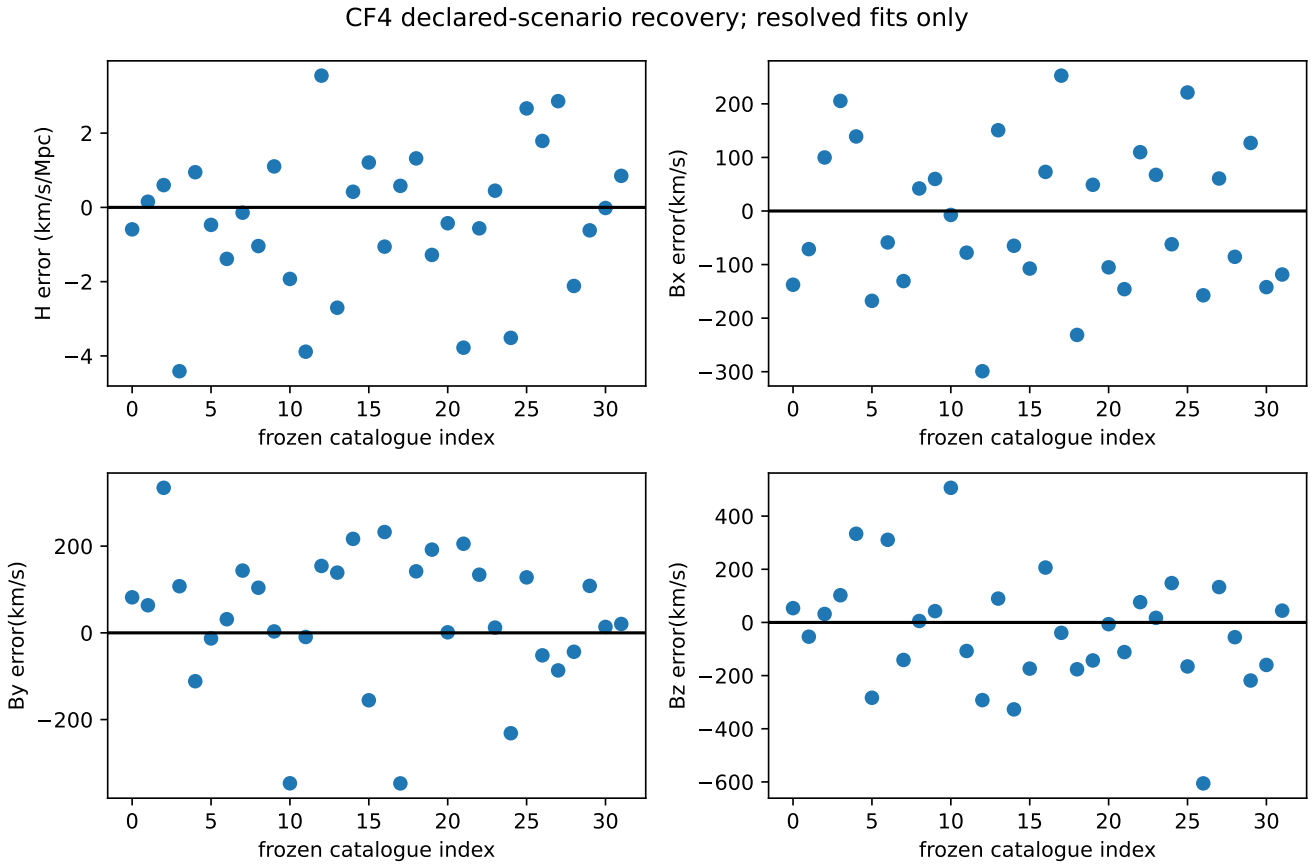


Figure 3: 32 conditional selected scenario injections, not empirical CF4 coverage

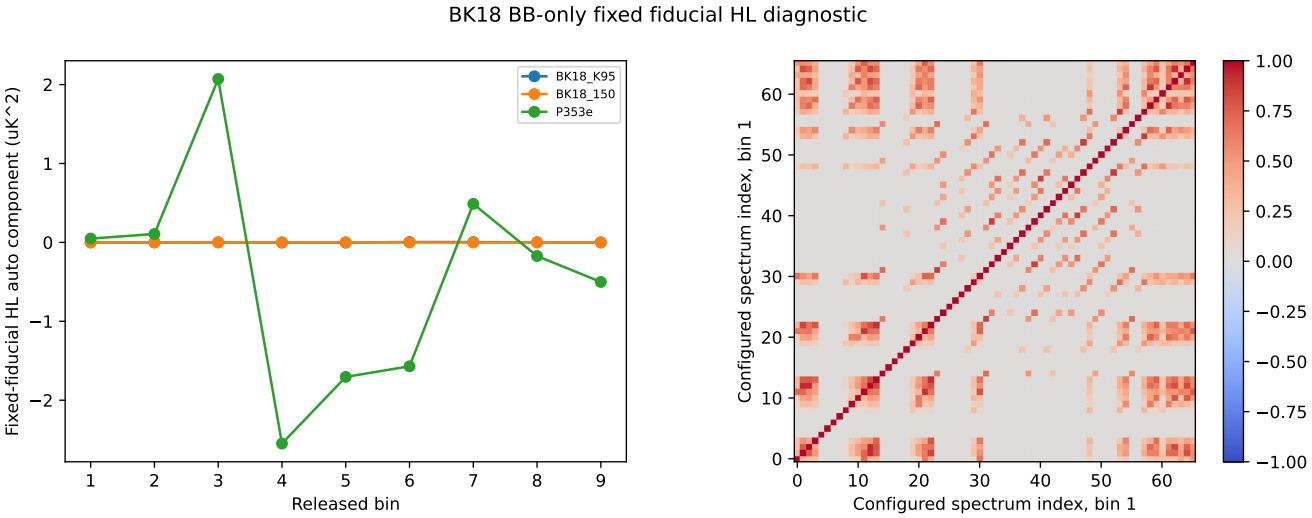


Figure 4: BK18 correlated fixed fiducial HL descriptor; no PTE or r fit

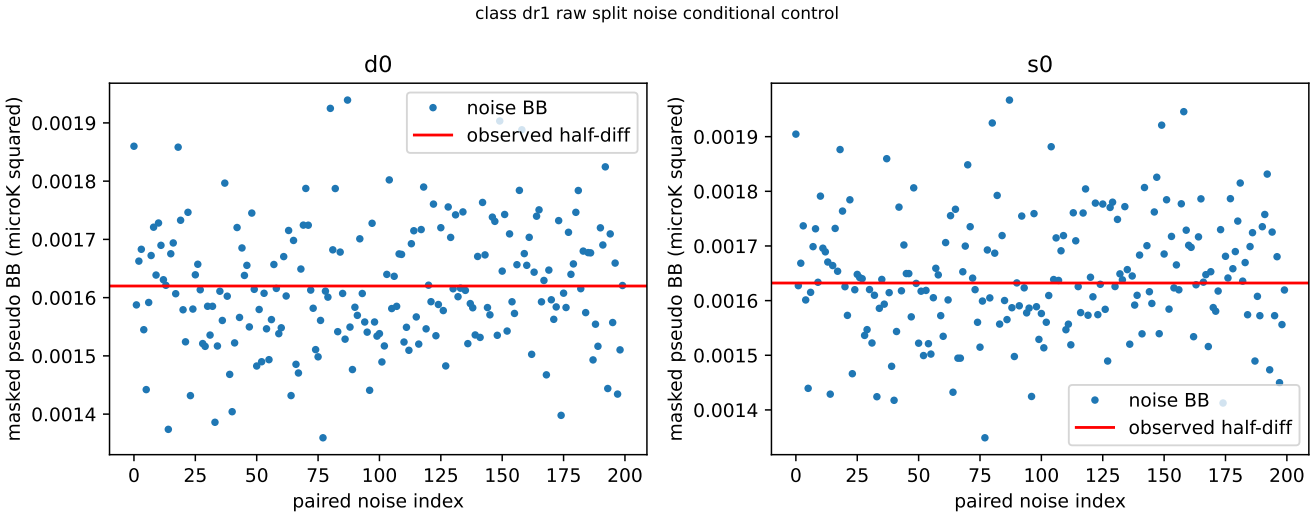


Figure 5: CLASS BB component only; joint six feature rank driven by EE; independent split-noise approximation

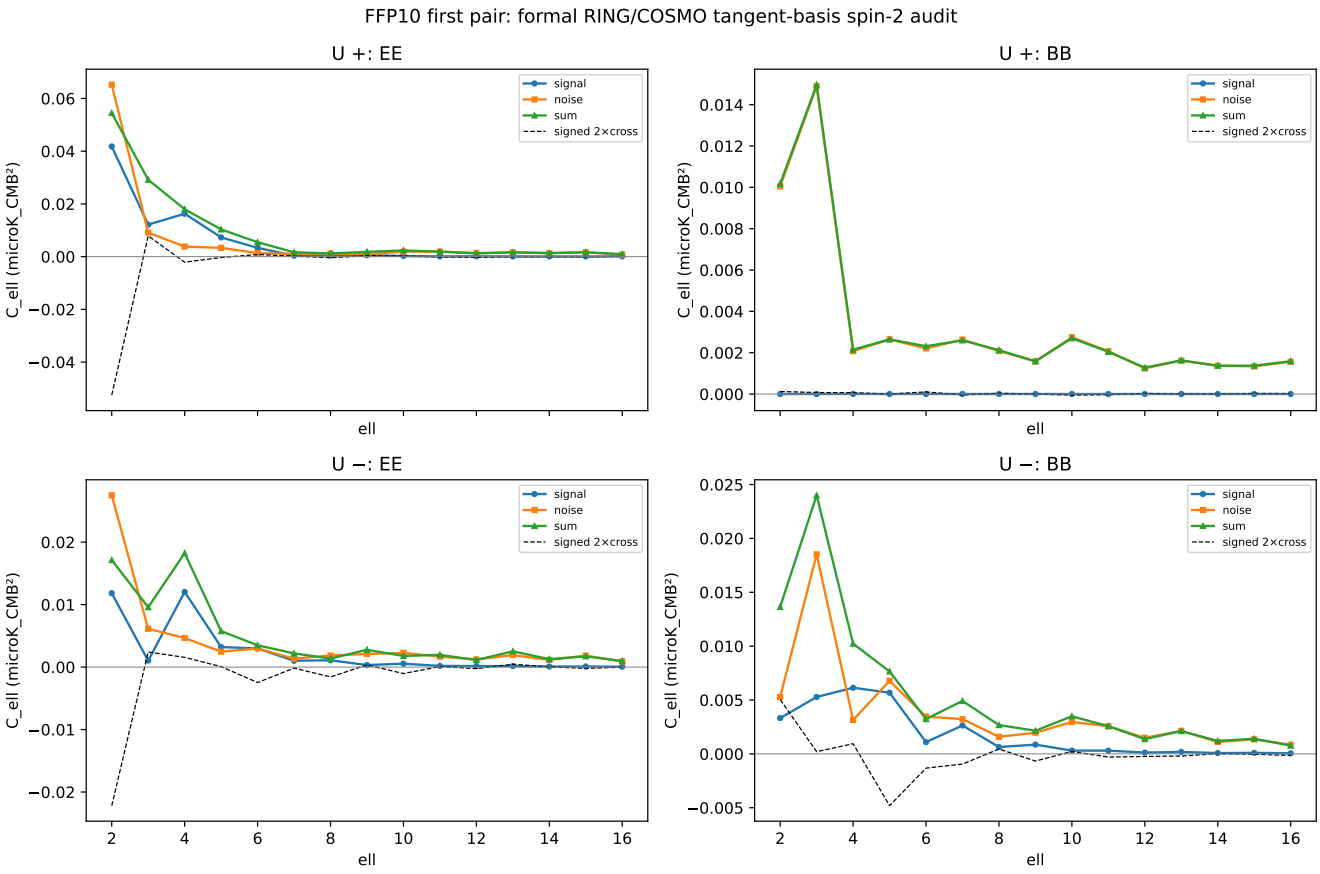


Figure 6: First raw pair formal spin-2 operator; U plus/minus are declared encodings; source bridge HOLD

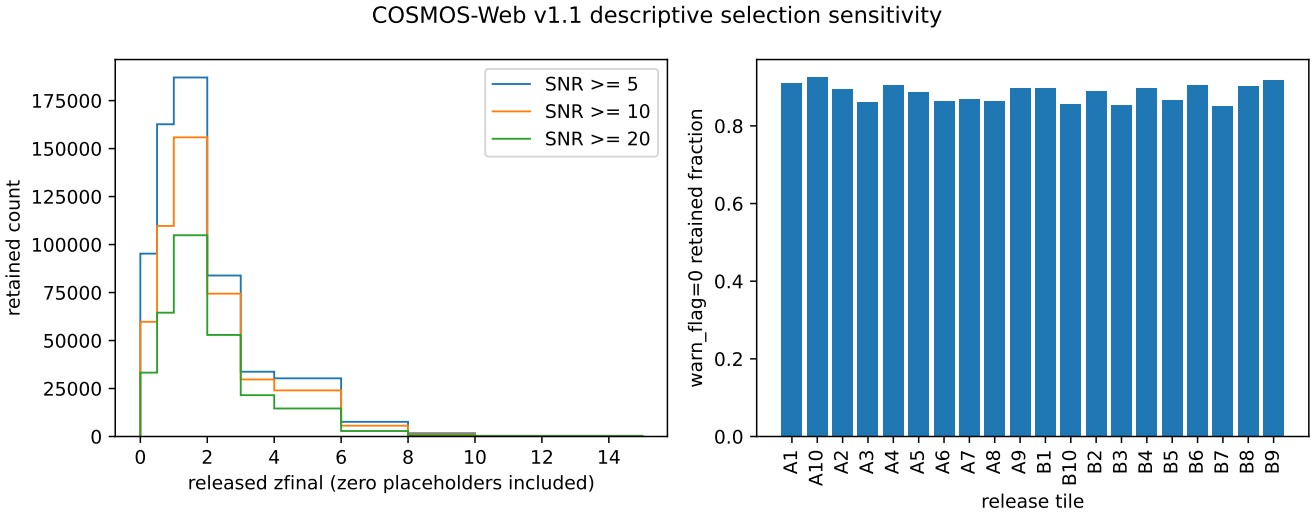


Figure 7: Released zfinal retention includes zero star/artifact placeholders; no galaxy population or completeness inference

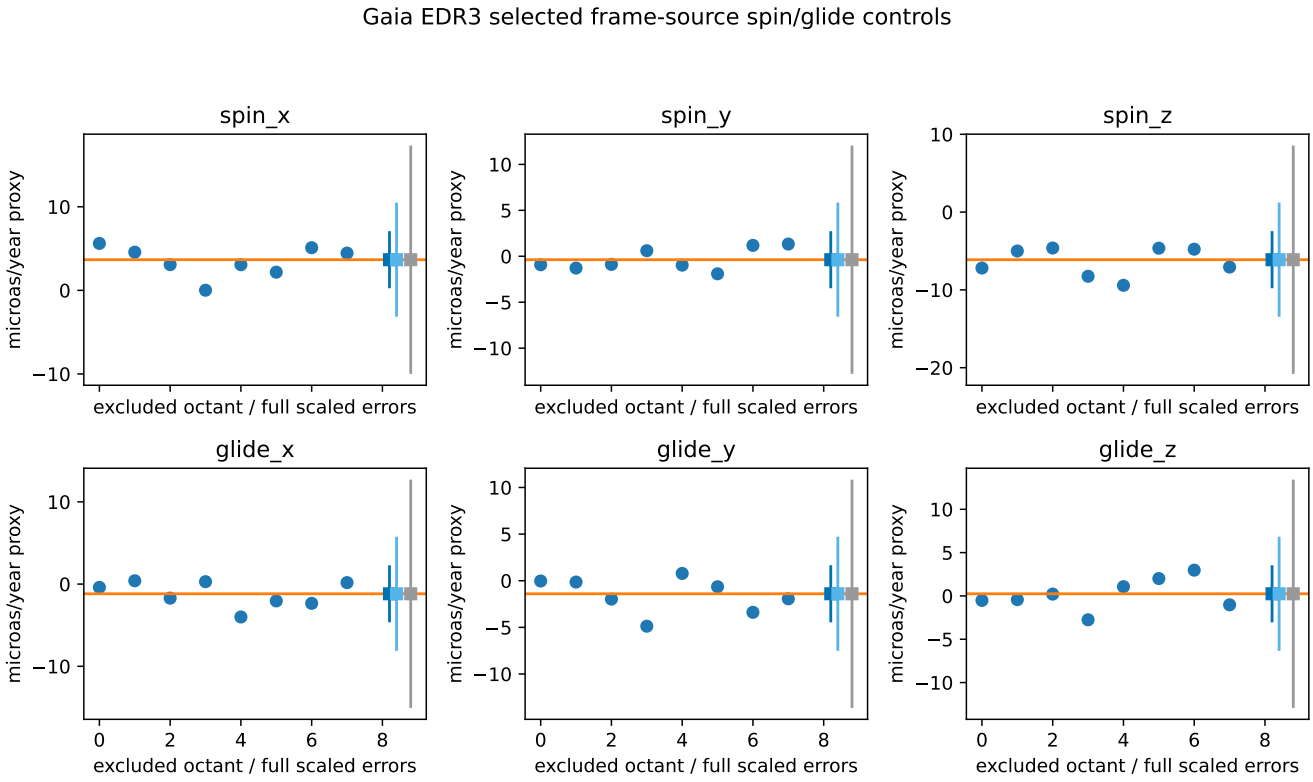


Figure 8: 4756 outcome-selected frame sources: spin/glide working model; no independent physics inference

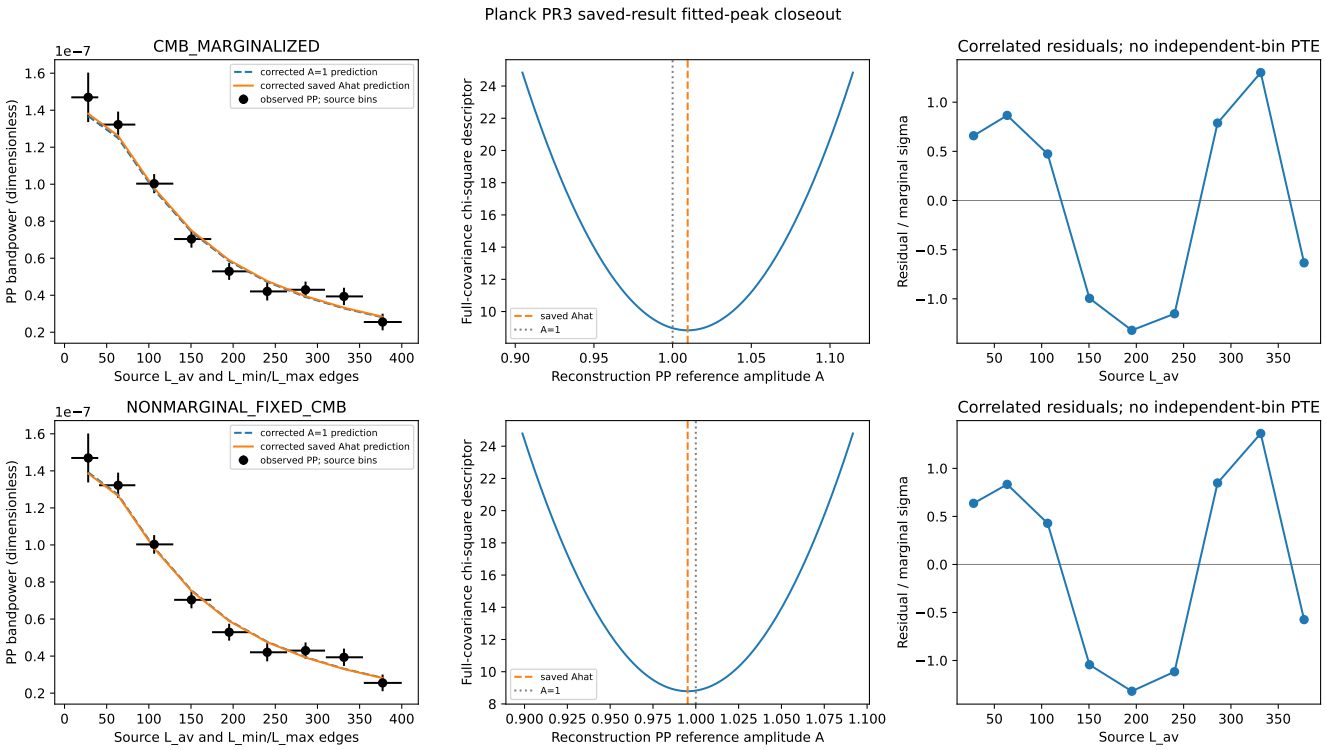


Figure 9: Planck PR3 source-corrected reconstruction amplitude; shared Planck information retained

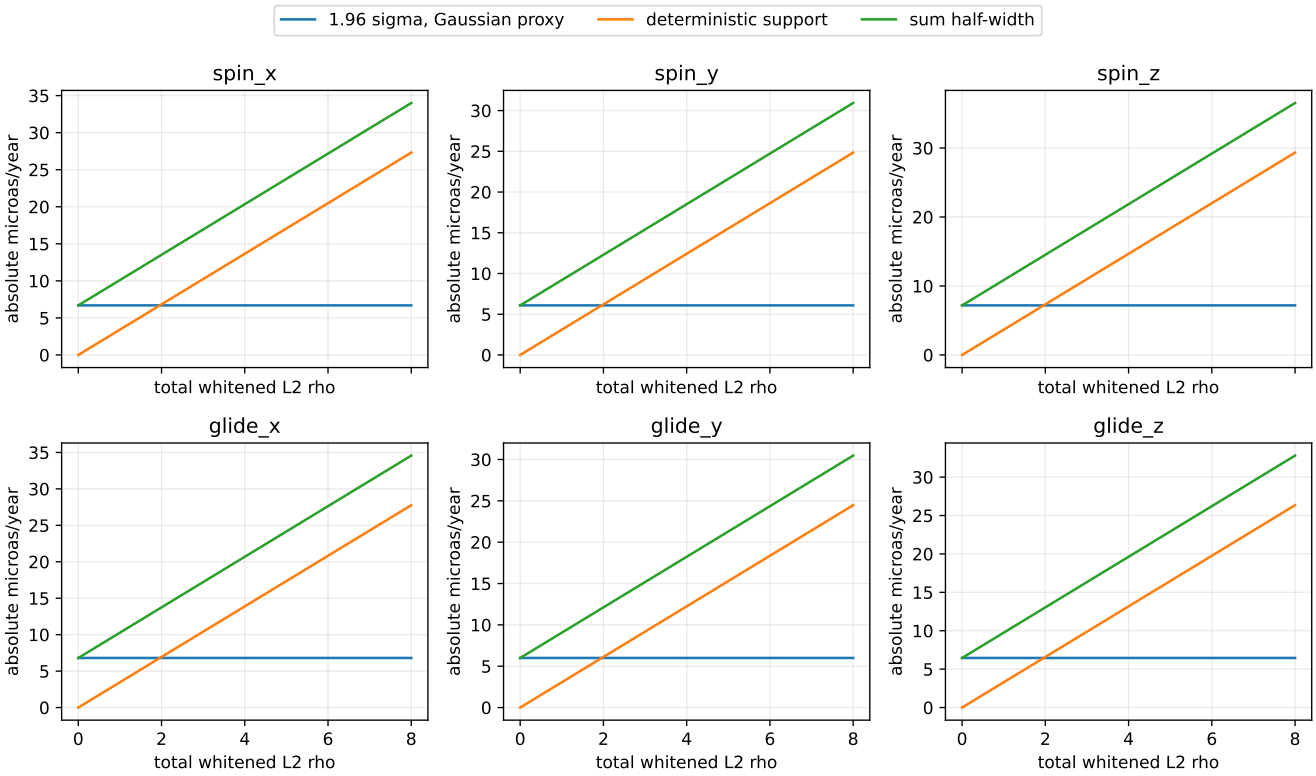


Figure 10: Gaia fixed-cohort Gaussian reference versus imposed TOTAL whitened-norm budget; coordinatewise only, no certified physical bound

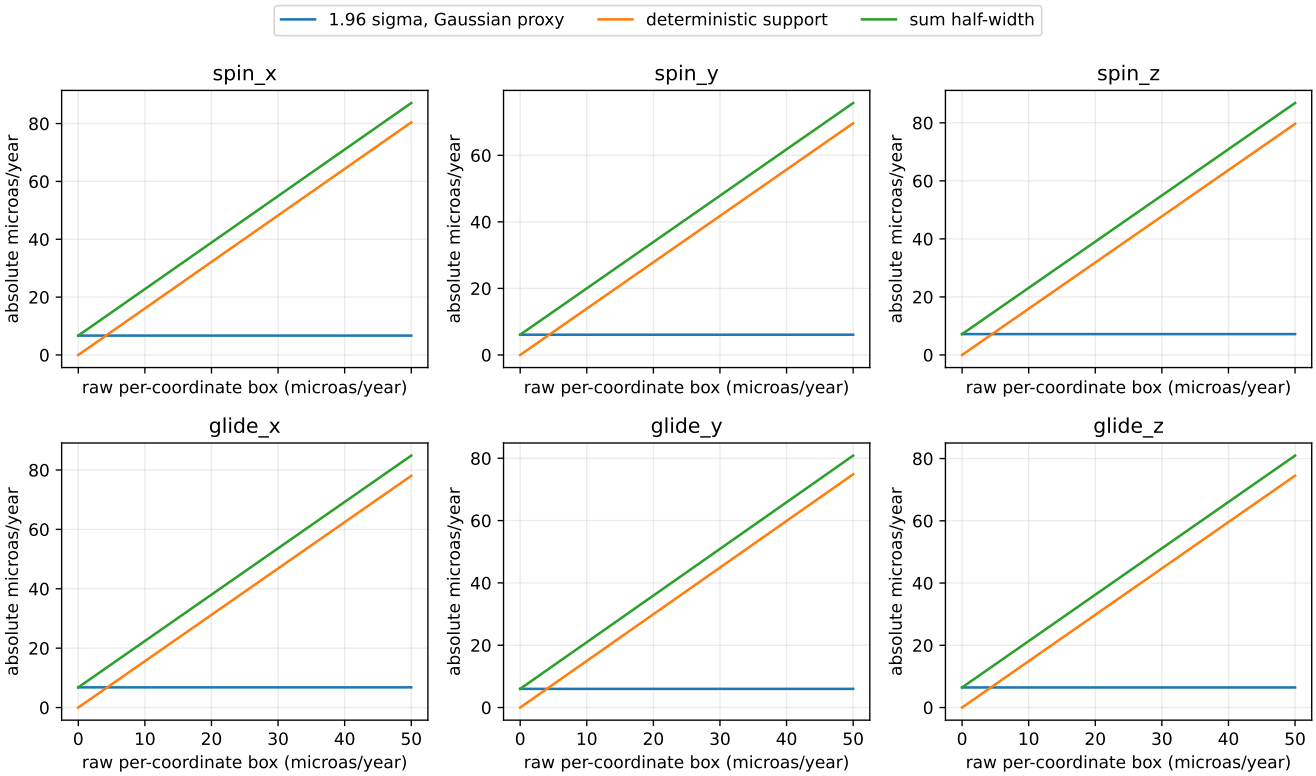


Figure 11: Gaia fixed raw tangent-coordinate box support; basis dependent, endogenous source selection, no empirical coverage